Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [14AE44]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2016

AERODYNAMICS-I

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions choosing at least **TWO** questions from each Part.
2. Any missing data may be suitably assumed

PART - A

1. a. State the integral form of C_l in terms of C_p for an inviscid flow and small angle of attack. **02**

b. Express lift and drag equations in terms of normal force (N), axial force (A) and angle of attack (α) for an airfoil with a neat sketch. **04**

c. Consider an airfoil with chord 'c' and the running distance 'x' measured along the chord. The LE is at (x/c)=0 and T.E at (x/c)=1. The pressure coefficient variation over upper and lower surfaces are given by. **08**

$$C_{pu}=1-300\left(\frac{x}{c}\right)^2 \quad for\ 0\le\frac{x}{c}\le 0.1$$

$$C_{pu}=-2.23+2.28(x/c) \quad for\ 0.1\le x/c\le 1.0$$

$$C_{pl}=1-0.95(x/c) \quad for\ 0\le(x/c)\le 1.0$$

Calculate lift coefficient for small angle of attack.

2. a. Define circulation (Γ) and vorticity (ξ).how they are related? **02**

b. Consider lifting flow over a cylinder of radius 0.5m. **04**

the free-stream velocity is 25m/s at standard altitude of 3km ($P=0.91kg/m^3$) and maximum velocity on the surface of the cylinder is 75m/sec. Calculate lift/unit span on the cylinder.

c. For a lifting flow over a cylinder get the lift equation in terms of circulation Γ (Kutta-Jaukowski relation). Draw sketches to show the flow fields and stagnation points for different Γ. **08**

3. a. How Kelvins circulation law is used to explain generation of circulation over an airfoil. **02**

b. Give the complex potential functions for **08**

i. Source flow

ii. Vortex flow and evaluate complex velocity of $W=6z^2$ @ point (6.4)

4. a. State the use of Blasius theorem. **02**

b. Give the summary of Kutta condition. **04**

c. Through Joukowski transformation, derive an expression for (t/c) ratio for symmetric airfoil. **08**

PART-B

5. a. Define aerodynamic centre (ac) and get an expression for (ac) terms of slopes of lift coefficient and moment coefficient. **04**

   b. Derive the basic equation of thin airfoil theory. **10**

6. a. State the assumptions made in lifting-line theory. **04**

   b. Derive the fundamental equation of Prandtl's lifting line theory. **10**

7. a. Define induced angle of attack and induced drag **04**

   b. An airplane crossing an altitude of 4km with $\alpha = 6^0$ has wing AR of 6.7 and taper ratio of 0.4. Calculate C_L and C_D of the wing (use airfoil data)assume e=0.99. **10**

8. a. Compare properties of boundary layer thickness and skin friction drag between laminar and turbulent boundary layers. **04**

   b. For the following airplane wing compare the total skin friction drag between a completely laminar flow and completely turbulent flow **10**

   Wing span=20 m; chord=3m

   Velocity= 200 m/sec; viscosity $\mu = 1.8 \times 10^{-5} kg/m-s$

   Altitude: standard sea level